



FSAE Rear Wing Design

RAHUL GUPTA

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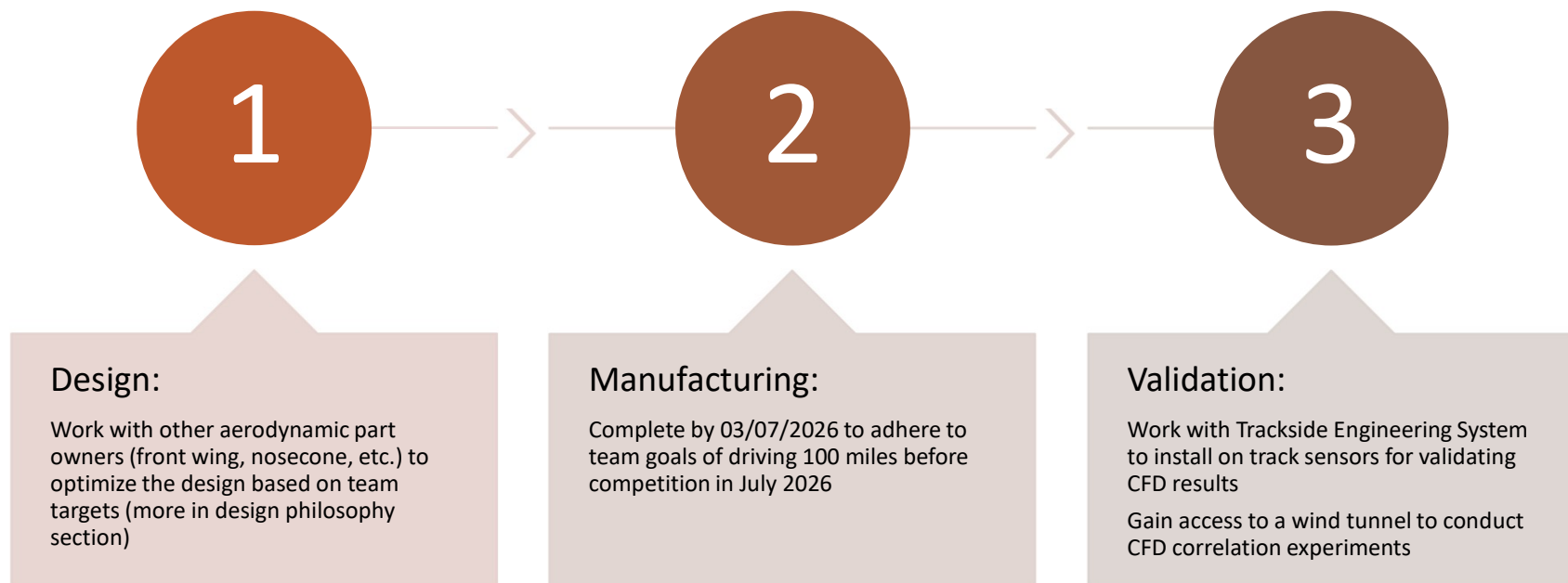
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Project Objectives

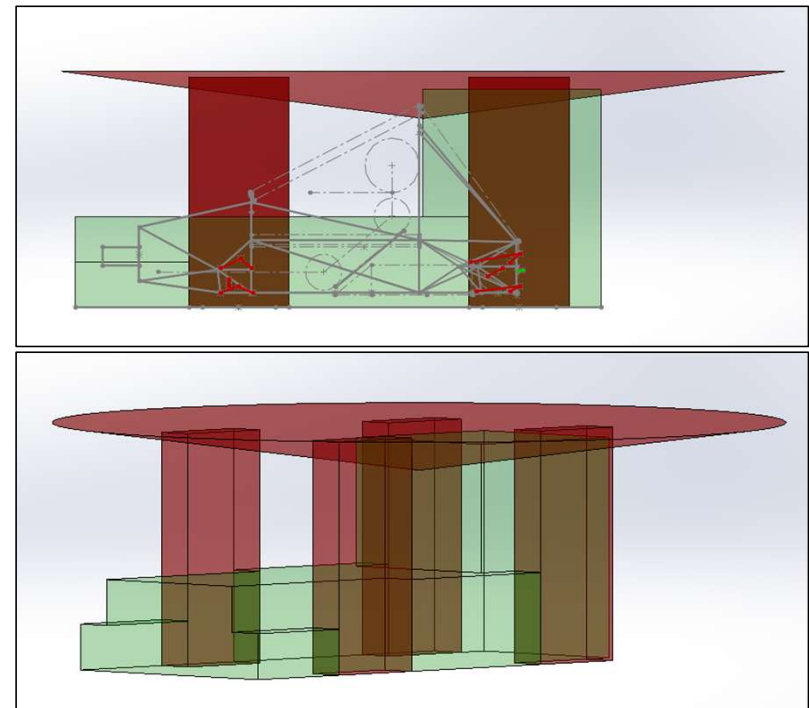


Requirements

Rule	Requirement
T.7.1.3	All aerodynamic devices must meet: 1. The mounting system provides sufficient rigidity in the static condition 2. The aerodynamic devices do not oscillate or move excessively when the vehicle is moving.
T.7.1.4	All forward facing edges that could contact pedestrians (wings, end plates, and undertrays must have a minimum radius of 5 mm for all horizontal edges and 3 mm for vertical edges
T.7.1.5	Other edges that a person may touch must not be sharp
T.7.5	In plan view, any part of the aerodynamic device must be: 1. No more than 700 mm forward of the fronts of the front tires 2. No more than 250 mm rearward of the rear of the rear tires
T.7.6.2	In plan view, any part of the aerodynamic device must be: When between the centerlines of the front and rear axles: Inboard of a line drawn connecting the outer surfaces of the front and rear tires at the height of the wheel centers
T.7.6.3	When rearward of the centerline of the rear wheel axles: In the Rear Aerodynamic Zone
T.7.7.1	Any part of any aerodynamic devices that is located: 1. In the rear aerodynamic zone must be no higher than 1200 mm above the ground 2. Outside of the rear aerodynamic zone must be no higher than 500 mm above the ground

Requirements

Rule	Requirement
IN.8.2.2	If any loaded deflection is significant, then a force approximately 200 N may be applied. 1. Loaded deflection should not be more than 25 mm 2. Any permanent deflection less than 5 mm
EV.5.10.3	Mounting location of each Ready to Move Light must: a. Be near the Main Hoop near the highest point of the vehicle b. Be inside the Rollover Protection Envelope F.1.13 c. Be no lower than 150 mm from the highest point of the Main Hoop d. Not let the driver's helmet touch it e. Be visible from 1300 mm vertically from ground level, inside a 2000 mm horizontal radius from the light <i>Visibility is checked with Bodywork and Aerodynamic Devices in place</i>
PTN	Fusebox must be accessible without removing rear wing



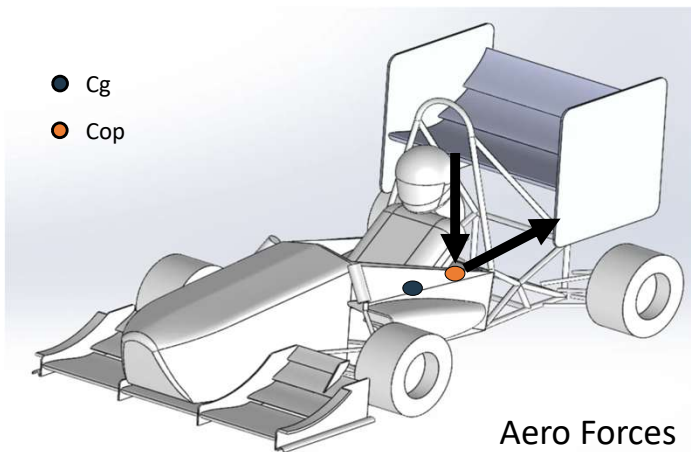
Design Philosophy

Aerodynamic Components

- Add downforce → Increase lateral acceleration → Increase cornering speed → Increase lap time
- Add drag → Negative work on vehicle motion → Decrease longitudinal acceleration → Decrease lap time
- Add weight → Increase overall car mass → Decrease longitudinal acceleration → Decrease lap time

Center of Pressure

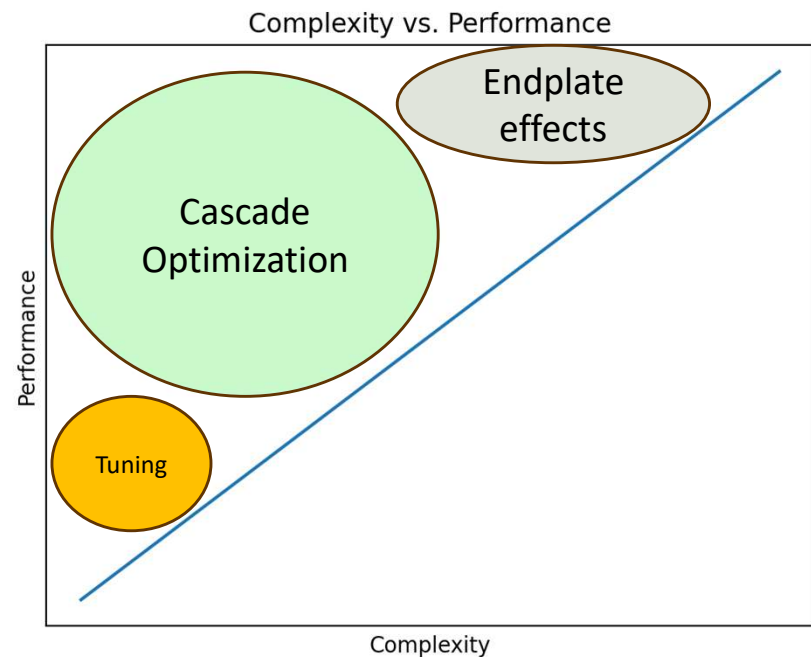
- Where all aerodynamic forces act
- Position relative to Cg affects overall handling



Design Philosophy

Aerodynamic Balancing

- Lift/Drag Balancing through Cascade Optimization
 - Base of the RW design
 - Easily manufacturable
- CoP balancing through Endplate effects
 - More complex features
 - Help adjust for overall aerodynamic package goals
- Performance balancing through Tuning
 - Simple adjustments (e.g., Low Drag Configuration for acceleration event)
 - Design for various car attitudes throughout laps (e.g., pitched forward going into a turn)



Design Philosophy

Airfoil Choice: S1223

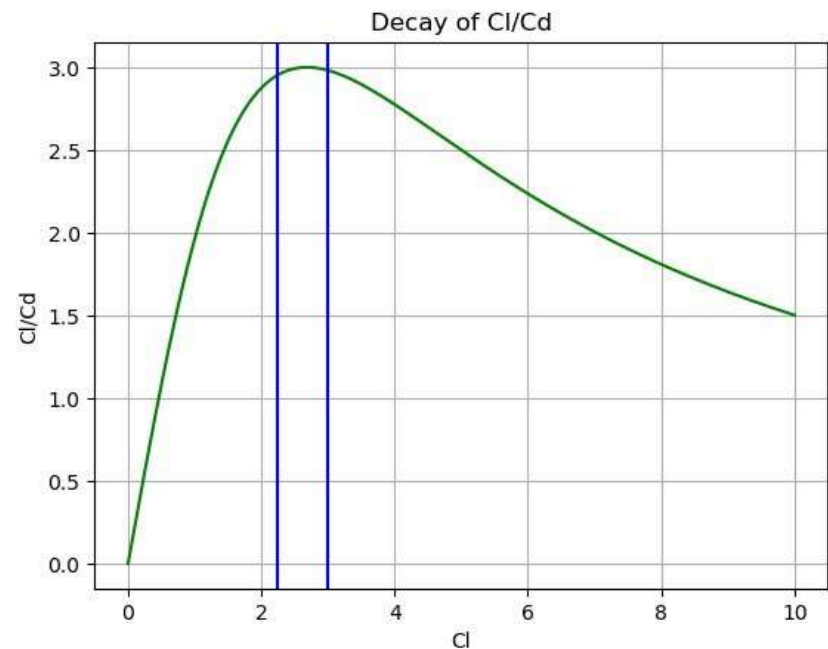
- Recommended by FSAE
- Aggressive airfoil for high downforce designs to achieve aero targets

Main Cascade: 3 Elements

- As opposed to a two-element design to hit higher C_l based on aero targets

Upper Cascade: 2 Elements

- Make use of the full bounding box to achieve aero targets
- Use freestream air unaffected by driver, headrest, and roll hoop

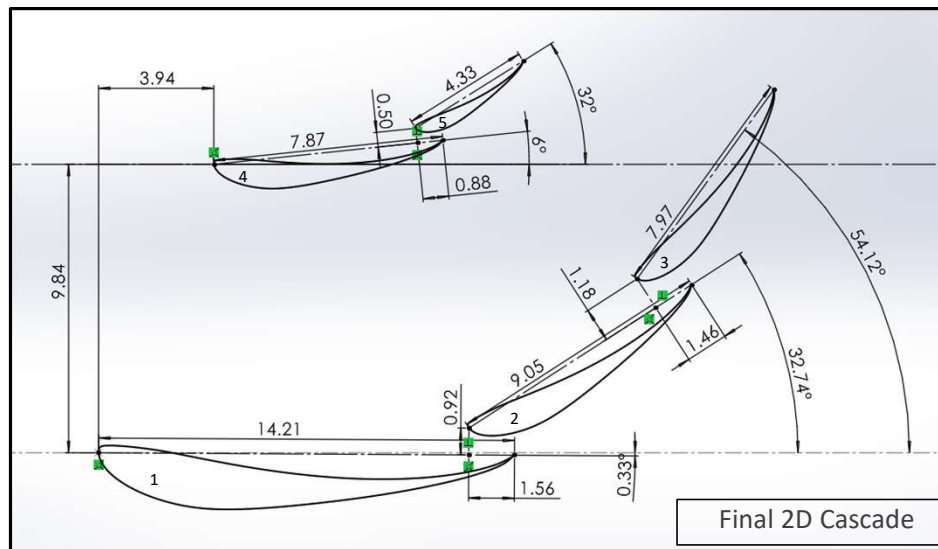


Aero targets justified through a C_l vs C_l/C_d graph

2D Parameterization Sweeps

Started with 2D CFD on the initial profile

- Simple meshing and solution process for fast iterative design
- Changed angle of attack (α) per element, gap length between elements, position of upper elements, and chord length of upper elements



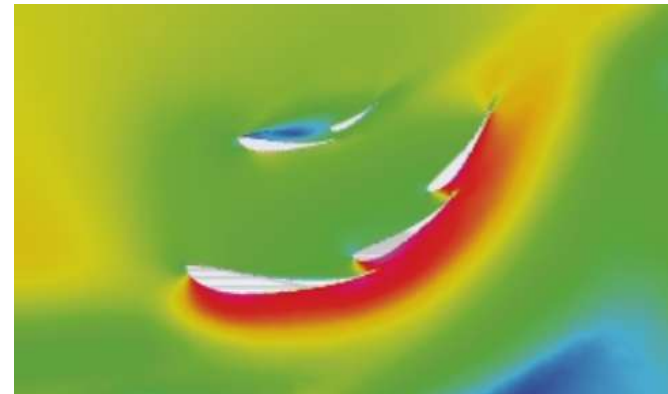
Parameters	CI*A
Pre-Parameterization	2.76
α_1, α_2 , Slot Gap 1-2	2.54
α_1, α_3 , Slot Gap 2-3	2.99
α_4 , 4th Chord, 4th Position	3.02
α_4, α_5 , 5th Chord, Slot Gap 4-5	3.36

3D CFD

- Extruded 2D profile with a wingspan of .9m to maximize bounding box usage
- Added basic square endplates to prevent vorticity generation on wingtips
- Slowly increased complexity of simulations, adding chassis, driver, headrest, nosecone, and front wing
- Saw downforce decrease due to non-freestream air and front wing wake

Comparison of RW velocity profile with free-stream air (left) vs non-free-stream air (right)

Both simulations take the same cut plane and have the same velocity magnitude color scale



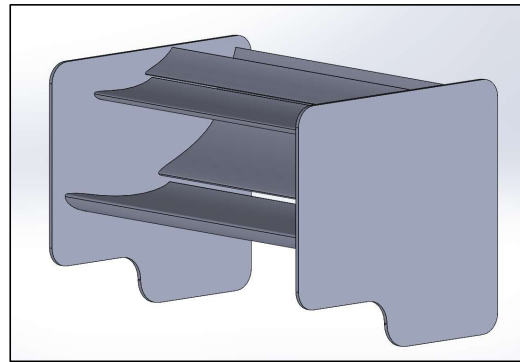
* Taken from simulations run on a previous iteration of the wing (upper element angle of attacks were changed after these simulations)

3D CFD

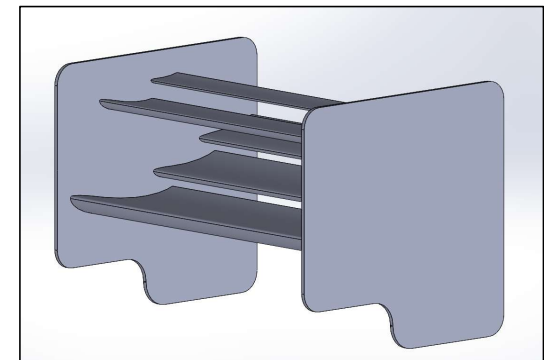
Tuning

- Created Low-Drag Configuration for acceleration and endurance events
- Ran simulations at various car attitudes to justify performance throughout a lap

Configuration	Cl*A	Cd*A
Static	2.20	0.83
Low-Drag	0.68	0.12
Forward Pitch (2°)	2.15	0.65
Backward Pitch (1°)	2.04	0.71
Roll (2°)	1.41	0.50
Yaw (15°)	1.85	0.79



Static RW (left) &
Low-Drag RW (below)



Manufacturing & Assembly



RW On Car

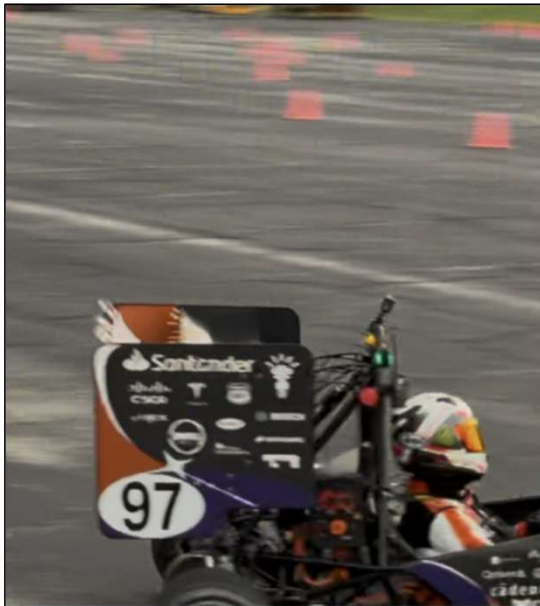
Airfoil Cascade

- Wire cut foam molds to airfoil profile and chord
- Wet layups over those molds, vacuum sealed and cured at ambient
- Main cascade uses a ribs (aluminum waterjet) and spars (carbon rod) assembly
- Upper elements use a foam core

Endplates

- Flat wet layups, vacuum sealed and cured at ambient
 - 6 total plies: two 0° , 45° , 0° sheets with a Nomex honeycomb core in between
- Waterjet holes with bonded machined inserts

On-Track Validation



Attached tufts during
endurance testing

Tuft Testing

- Applied tufts across the rear-wing elements to visualize local flow direction during on-track running
- Photos and video showed flow attachment throughout laps

Trackside Tuning

- Driver feedback identified oversteer during endurance-focused testing
- Added a Gurney flap to the third element to increase rear downforce improve aero balance

Next Steps

- Integrate PPS Aerostrips from our recent sponsorship onto the rear wing
- Evaluate placement using tuft behavior, driver feedback, and endurance consistency

Areas of Improvement

Front-to-Rear Wing Interaction

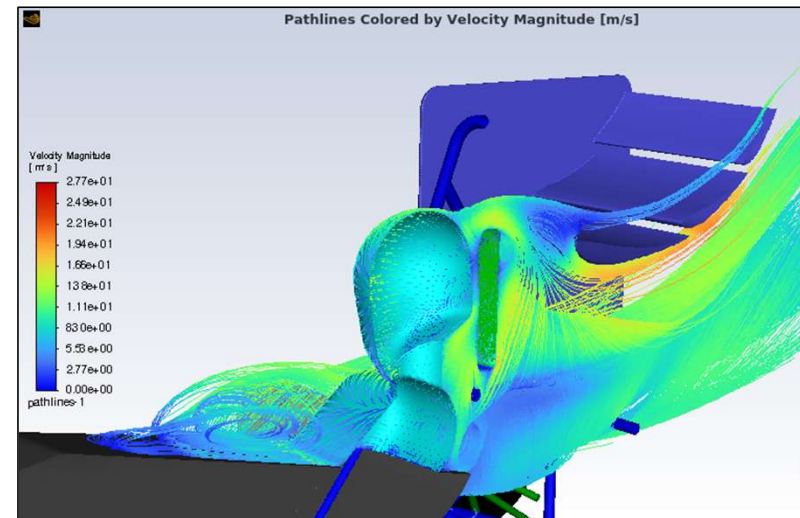
- The longitudinal region inboard of tires but outboard of nosecone sees increased front wing upwash (see contour plot on slide 10)
- Possible Solutions: Inverted Airfoils, Decreased Front Wing Camber

Spanwise-Curved Main Element

- The center section sees a different effective angle of attack behind the driver
- Curve the main element to tailor local incidence instead of using one straight geometry across the full span

Endplate Complexity

- Use more complex endplate geometry to weaken tip vortices and reduce induced drag (louvres, cutouts)
- Evaluate small diveplanes for additional downforce without excessive drag or flow separation



Driver/headrest interaction